

# Optimality: exhaustive permutation enumeration vs SciPy Hungarian

instances ( $n = 1..7$  square, plus rectangular reserve/unmet cases) match exactly in rational

instance count for  $n = 7$  enumeration: 5040 permutations. Ties documented separately (ASSIGNMENT-TIE-001).

exactly verified on tested domain ( $n \leq 7$ )

Wolfram 15.0.0 | WorkingPrecision 50 | cross-language tol  $1e-9$  |  $n = 24$  instances, exact arithmetic

